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Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES
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Professional Installation Manual Ubuntu Server LTS Deployment

Prepared by: Sandoval, Geoff

Submitted to: John Peñaredondo, MIT



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1. Introduction

This manual provides a structured, repeatable procedure for installing and configuring Ubuntu Server LTS in a virtual machine. It is written so that another system administrator can follow it independently, without requiring additional verbal instructions, and arrive at the same working server configuration.

2. Ubuntu Server Installation Guide

2.1 Prepare the Virtual Machine

Setting	Value
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 virtual processors
Storage	40 GB
Disk	VDI or VMDK
Network	NAT
ISO	Ubuntu Server LTS

2.2 Begin Installation

Start the virtual machine and boot from the Ubuntu Server ISO. Select **English** as the installation language and configure the appropriate keyboard layout.

2.3 Configure Network

Use DHCP. Record the assigned IP address for later verification.

2.4 Configure Hostname

Set the hostname to **server01**.

2.5 Create User

Create the administrative user required for the server, using a strong password. The password is never recorded in any documentation.

2.6 Configure Storage

Select **Guided - Use Entire Disk** and choose the 40 GB virtual disk. Record the resulting partition scheme, file system, and disk size.



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2.7 Install SSH

Select **Install OpenSSH Server** and complete the installation.

2.8 Restart

Restart the server once installation finishes and remove the installation media.

3. Initial Server Configuration

After logging in for the first time, run the following commands in sequence to confirm the server is configured and reachable.

Verify the hostname

```
hostname
```

Verify the network

```
ip addr
```

Test internet access

```
ping -c 4 google.com
```

Update the operating system

```
sudo apt update  
sudo apt upgrade -y
```

Check SSH

```
systemctl status ssh
```

The service should report **active (running)**.



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4. Server Verification Procedures

Verification	Command	Expected Result
Hostname	hostname	server01
IP Address	ip addr	Assigned IP address shown
Internet	ping -c 4 google.com	Successful replies, 0% packet loss
Updates	sudo apt update	Repository information retrieved
Upgrade	sudo apt upgrade -y	Packages updated
SSH	systemctl status ssh	active (running)

5. Troubleshooting Guide

Problem: No Internet Connection

Possible cause: Incorrect virtual network adapter configuration.

Recommended check: Confirm the VM network adapter is set to NAT (or the instructed mode) and that DHCP has assigned an address. Restart networking with `sudo systemctl restart networking` if needed, or recheck the adapter settings in the hypervisor.

Problem: SSH Is Not Running

Check the service status:

```
systemctl status ssh
```

If SSH was not installed during setup, install it with:

```
sudo apt install openssh-server -y  
sudo systemctl enable --now ssh
```

Problem: Incorrect Hostname

Check the current hostname:

```
hostname
```



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If it does not read **server01**, update it with:

```
sudo hostnamectl set-hostname server01
```

Problem: Cannot Identify the IP Address

Run:

```
ip addr
```

Check the active network interface (commonly enp0s3 or eth0) for the assigned inet address.

6. Best Practices

- Use strong, unique passwords for every administrative account.
- Never include passwords in screenshots or documentation.
- Keep the operating system updated with regular patch cycles.
- Document configuration changes as they are made, not after the fact.
- Keep screenshots organized in a dedicated folder with clear, sequential filenames.
- Use meaningful, descriptive filenames for all deliverables.
- Verify services are running after every installation or configuration change.
- Maintain backups of important configuration files and documentation.
- Use version control (Git) for portfolio files.
- Do not modify an existing VM when a separate, dedicated VM is required for the task.

7. References

Ubuntu Documentation Team. *Ubuntu Server Guide*. Canonical Ltd.
<https://ubuntu.com/server/docs>